

KYC process in Europe: challenges and opportunities

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Summary

- Recent regulations (4.1 AMLD) enforce EU banks to perform more KYC controls, collect more data and pay higher fees for mistake. At the same time General Data Protection Regulation (GDPR) threatens banks with conflicting and challenging data collection and protection requirements
- Current KYC approaches lack precision and reliability, every year process becomes more expensive and time-consuming
- Fragmented systems and platforms limit the ability to automate transaction monitoring and due diligence. Instead, compliance teams spend the bulk of their time collecting data, and then on “stare and compare” sessions, instead of investigative work
- Leading banks are trying to crack KYC problems by turning to new technologies
 - KYC solutions based on centralized ledgers are widely adopted since 2014
 - In 2018 first solution built on private blockchain (R3 Corda) was tested in large scale. It solves most of the KYC pain points, is trustless, allows controlling own data and supports gradual network adoption
 - Early-stage solutions on public blockchain are being built, but none is widely adopted yet
- More blockchain enabled KYC solutions could be on the horizon, but it requires significant collaboration across industry participants and a number of hurdles to overcome

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KYC is a critical step in anti-money laundering (AML) compliance obligatory in EU

Stakeholder	Goals in KYC process
Regulators	• identify customers perpetrating financial crimes (theft, fraud, money laundering and terrorist financing)
Financial institutions	• meet regulatory requirements • cut down own fraud losses • improve overall customer relationships
Customers	• receive access to financial services • perceive control over sensitive personal data

All parties involved (regulators, banks, customers) have to cooperate while aiming for conflicting targets in KYC process

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4th EU Money Laundering Directive 2015 (4.1 AMLD), Jan 2017 enforces banks to perform more KYC controls, collect more data and pay higher fees for mistake

1. Enhanced due diligence requirements regarding high-risk third countries

- Fewer scenarios where Simplified Due Diligence is acceptable
- All the institutions affected will be in need of reviewing their definition of high risk

2. Amplified definition and obligations regarding politically exposed persons (PEPS) and their families

- Stronger KYC Controls Required
- Definition of a PEP extended to include citizens holding prominent positions in their home country such as politicians, the judiciary and senior members of the armed services.
- Parents, spouses (or equivalent partners), children and their spouses or partners are also PEPs

3. Expanded reach to cover more types of organizations

- Include fintech companies such as virtual currency wallet providers and exchangers

4. E-money products will be specifically regulated under MLD4 for the first time.

- Virtual currency exchange platforms are now included and must go through the same level of scrutiny than traditional financial institutions

5. KYC thresholds have been lowered

- Limits that were previously set at €2500 are now at €250

6. The access of Financial Intelligence Units (FIUs)

- From January, 2017 FIUs will be allowed to access to centralized bank and payment account registers or electronic data retrieval systems

7. Sanctions have been updated

- The max fine has been set at least twice the amount of the benefit derived from the breach or at least €1 million
- Breaches committed by credit or financial institutions will cost at least €5 million or 10% of total annual turnover
- Natural persons involved with these breaches will be penalized with a maximum fine of at least €5 million

Stricter regulations will result in higher costs for banks and longer waiting time for customers

Providers of exchange services between virtual and fiat currencies, custodian wallets and most ICO projects in EU also have to comply with the KYC directives

Subject to KYC

Free of compliance

Exchanges

Exchanges offering virtual currencies to fiat exchange as a service

If exchange between two parties is not a service or exchange of crypto to crypto

Wallets

If the private key to the wallet is held by the wallet provider and the funds in the wallet belong to the user and not the wallet provider
Example: Bitwala (or Coinbase), which stores the private key (even though they might share it with the users)

If the user holds the private key to the wallet, and none of third party providers also hold it, Example: using MyEtherWallet (or Ledger Nano S, Trezor, Jaxx, Mist) to open a wallet on a blockchain network

ICOs

Not obliged to KYC when raising funds, (may do it voluntarily)

Obliged to KYC when exchanging the raised cryptos to fiat and deposit them at a bank account for their operational needs

As of now all crypto exchanges, custodian wallets and most ICOs need to incorporate own solutions for KYC

Since May 2018 General Data Protection Regulation (GDPR) threatens banks with conflicting and challenging data collection and protection requirements

Under GDPR, banks and other organisations who carry out identity checks and hold sensitive information about customers will have to be completely transparent about what happens to that data after it has been used.

GDPR covers the following rights for individuals:

- the right to be informed
- the right of access
- the right to rectification
- the right to erasure ("right to be forgotten")
- the right to restrict processing
- the right to data portability
- the right to object
- the right not to be subject to automated decision-making including profiling

GDPR

- Clients can demand the data held on them and/or erase such data from their systems
- Penalties for banks are significantly tougher than for KYC non-compliance: up to €20 million or 4% of global turnover

KYC

- Requires the collection and retention of a large volume of client
- Regulators can demand this data within 72 hours of a request

Technology in the data protection sphere will need to evolve to manage the conflict between GDPR and KYC

KYC process

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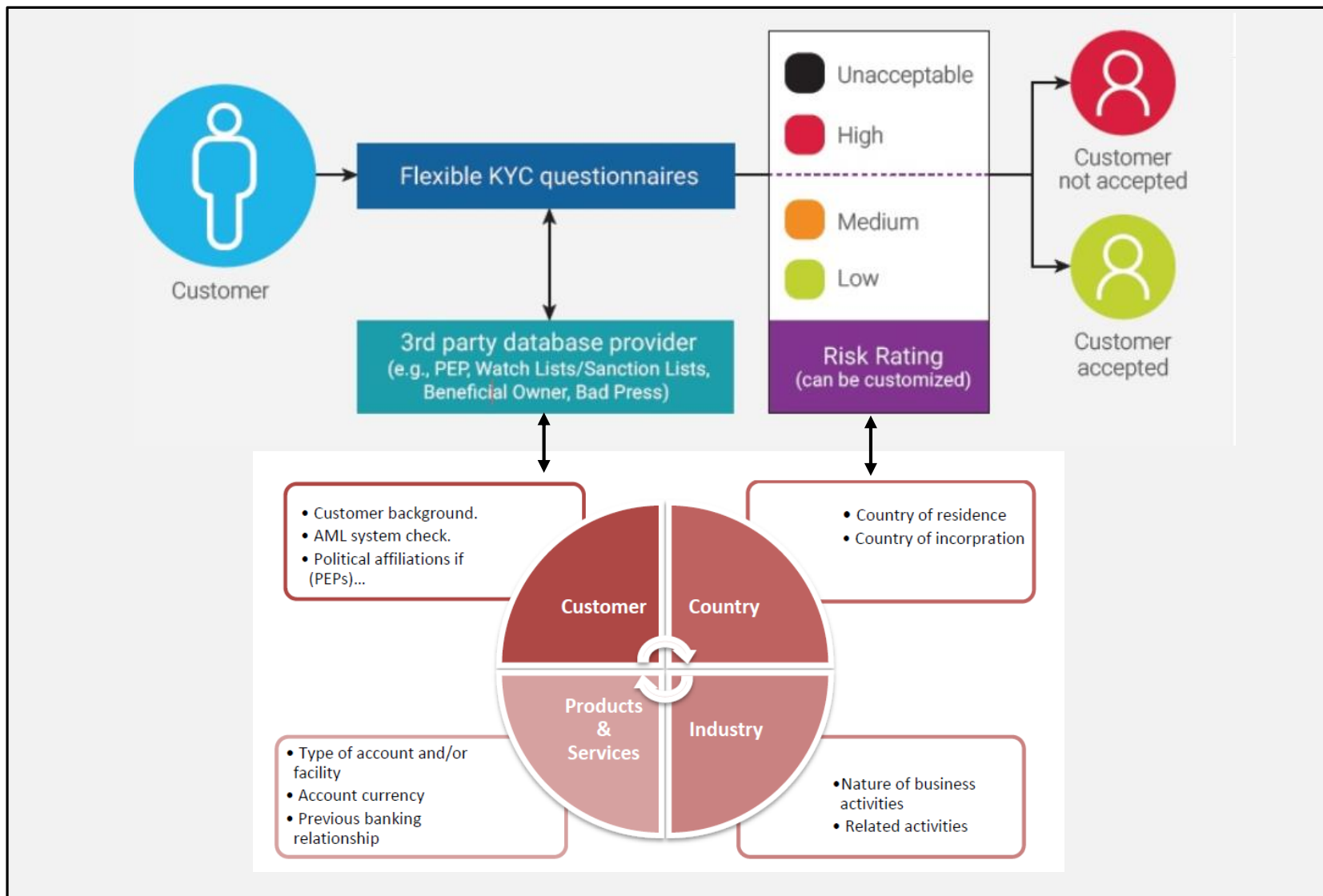
Challenges

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Typical KYC process involves multiple re-occurring controls and iterations

Step	Description	Functionality	Trigger
Customer onboarding	Information gathering during customer on-boarding to aid in the decision of whether or not an account should be opened	Questionnaire, risk scoring system	• Customer request for services
Re-review risk scoring	Deep inside to capture all out-of-the-box risk factors and change log information to re-trigger KYC assessments	Configurability of adding change log and re-trigger KYC for the newly added risk factors	• Change in risk score of any values of the risk factors/parameters • New value is added due to change in regulation (e.g. customers whose country of taxation is XYZ needs to be re-reviewed for regulatory or external factors, etc.)
Annual updates	Repetative rigorous tests on major clients at least every 12 months	Configurability of updating logs and all the systems relying on it	• Every 12 months

Case study 1. KYC for private customer onboarding with Emirates NBD, UAE



Case study 1. KYC data is being validate against hundreds of 3rd party data providers



Case study 2. KYC for legal entity customer with Barclays, UK

We will require personal information (name, ID, residence) for each official holding a Key Account Party, C-Suite or equivalent position, also for individuals with a significant controlling stake or ownership stake in the businesses banking with us (normally 25% equity)

We will ask you to describe structure of your business incl.

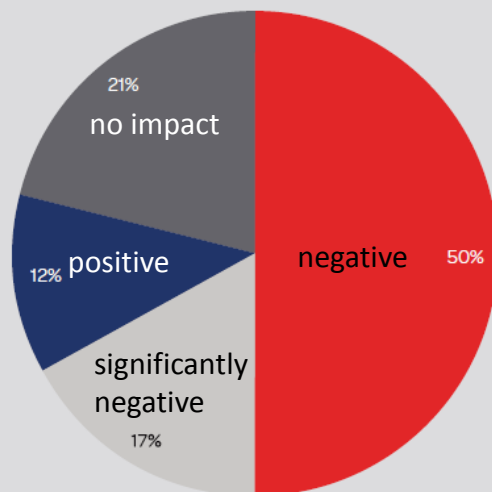
- intermediary companies
- country of registration and registered number
- the full breakdown of all share capital, including all share classes (this should total 100%)
- details of any bearer instrument/certificate which forms part of your share capital, and where this does not exist today, confirmation of whether or not the firm has the ability to issue share capital in bearer form
- details of voting control where this does not align to equity stake

28 - 45
days

We will regularly review the identification and verification documentation we hold on file for you – we usually refer to this as **Refresh**.

Sometimes it may seem like we're asking you for things that you feel we already know, you think aren't relevant, or that other banks do not ask for

67% negative or significantly negative impact of KYC on customer acquisition



“84% small companies are saying they did not have a good experience of KYC and for 27% onboarding took two months or more”

Thomson Reuters

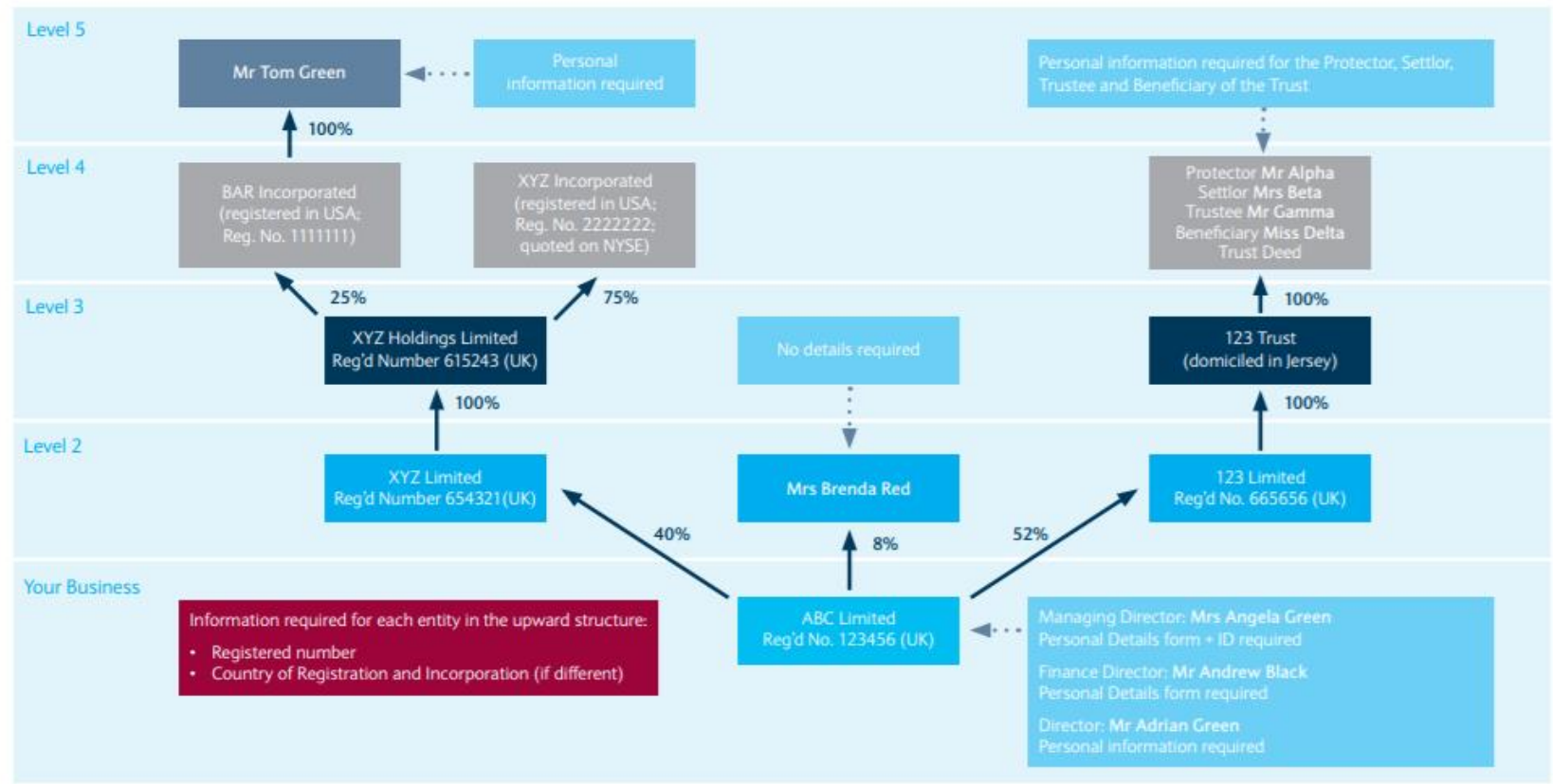
“Banks estimate that approximately 90% of our time is spent on collecting documents and 10% on actually analyzing them”

Fund Channel

KYC for legal entity customer with Barclays, UK is a becoming increasingly more complex due to evaluation of business structure and multiple stakeholders

- Bank **verifies information from data sources in the public domain**: lists of politically exposed persons (PEPs), “Watch Lists” of Foreign Assets Control (OFAC), HMRC Financial Sanctions list
- Where this is unavailable, **independent verification is required** by in-house legal team/accountant, external professional advisor, e.g. Solicitor/Lawyer
- **Corporates need to provide this information on average to 8-11 global banking relationships**

Example structure chart



KYC with a decentralized shipping and cargo platform BitNautic, Switzerland

Onboarding KYC requirements

- Ethereum wallet address
- e-mail address verification
- proof of identity (submitting an government or bank issues identity document with a photo)
- proof of residency (utility bill or bank statement)

Ongoing KYC controls

- all login attempts and IP addresses from users
- multiple accounts registered by the same user by comparing names
- addresses and IP addresses
- trading/Investment history
- Deposit
- withdrawal address from all users
- incoming transactions to all wallets

KYC compliance

- all KYC requirements on BitNautic are in sync with the laws of UK and Switzerland

GDPR compliance

- all our data is fully encrypted at rest and in transit, and we employ state-of-the-art firewall and backup technology
- all our data is continuously backed up in several high-security access-controlled data centers in different locations

Challenges

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Current KYC approaches lack precision and reliability

Scope of damage

- Global money laundering transactions, some of which reportedly fund terrorist groups, are estimated to be as high as **5% of global GDP, or roughly \$1-2 trillion annually**

False negatives

- Meanwhile, according to the United Nations Office on Drugs and Crime, **less than 1% of global illicit financial flows** are currently being seized by authorities

False positives

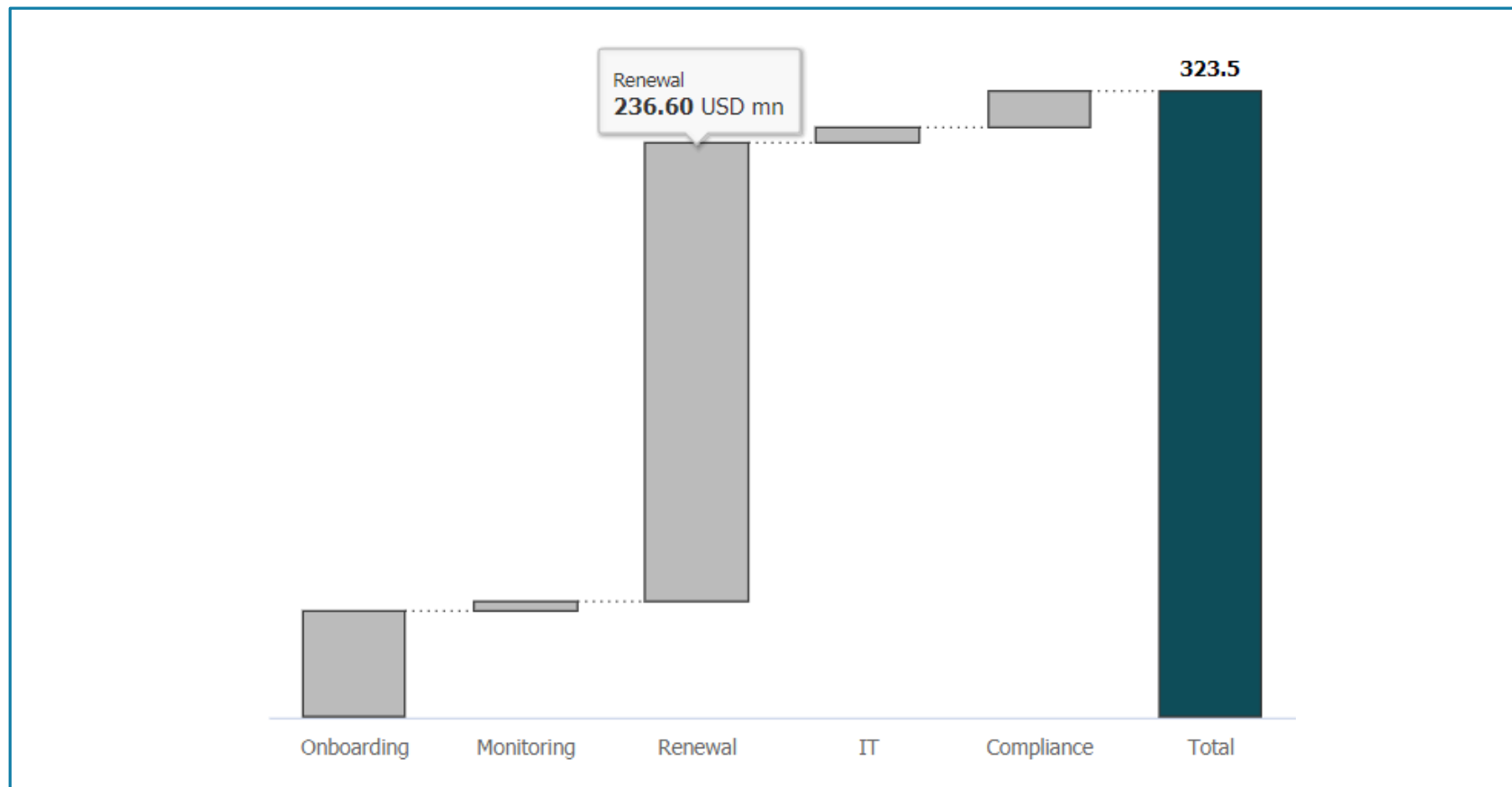
- Damage to client relationships from false positive alarms leaves 89% customers not satisfied with their KYC experience, and 13% changing their financial institution relationship as a result

Analytical approaches for customer risk scoring and transaction monitoring suffer from high rates of false positives, resulting in **significant resources focused on investigating lowrisk accounts and transactions.**

Adding new calibration tools and thresholds often leads to another spike in the number of false alerts.

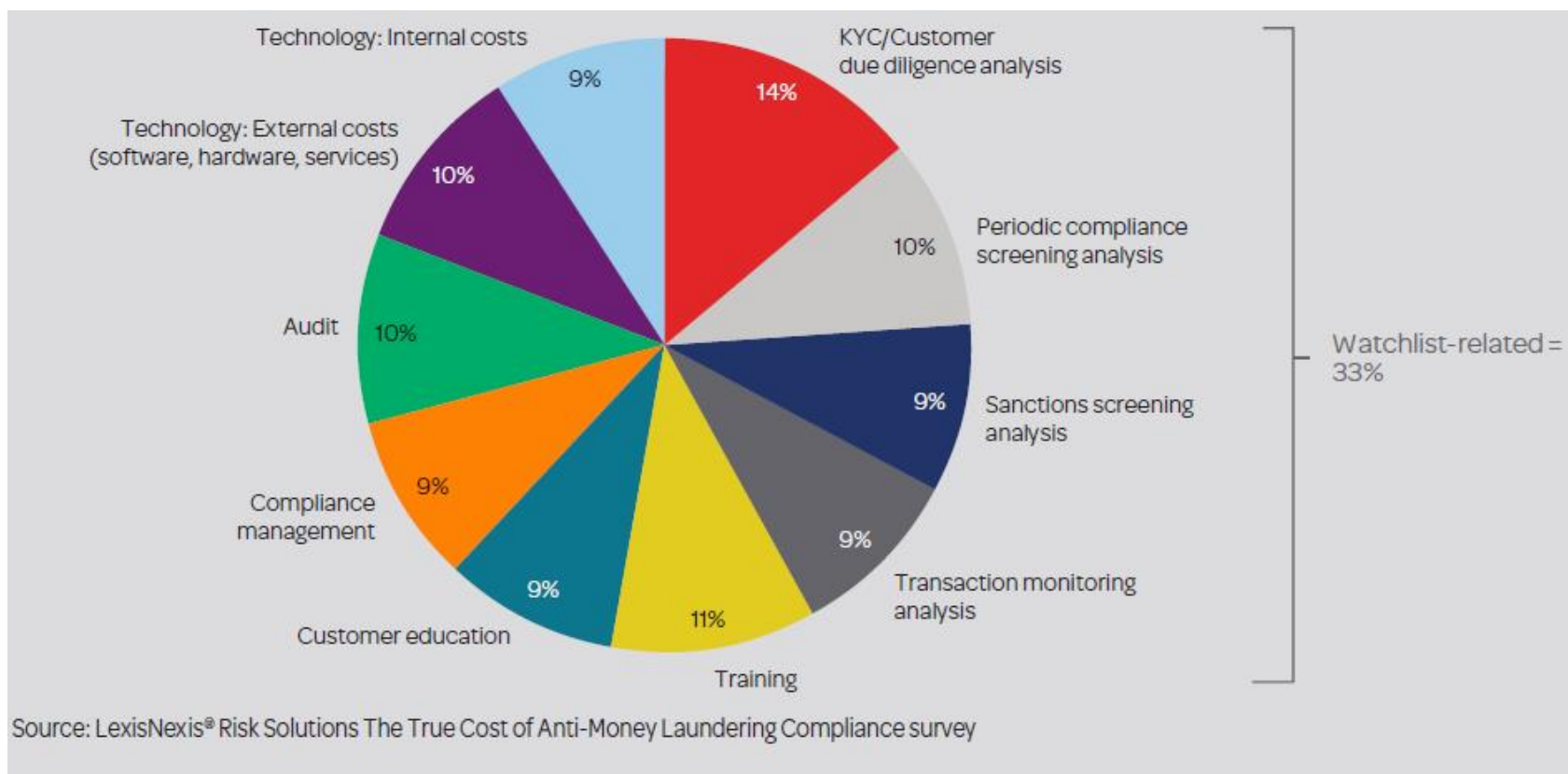
And at the same time KYC process is expensive and time-consuming

KYC costs in Sweden reached USD ~325mn p.a. and are mainly driven by time investment of personnel on the stage of customer records renewal



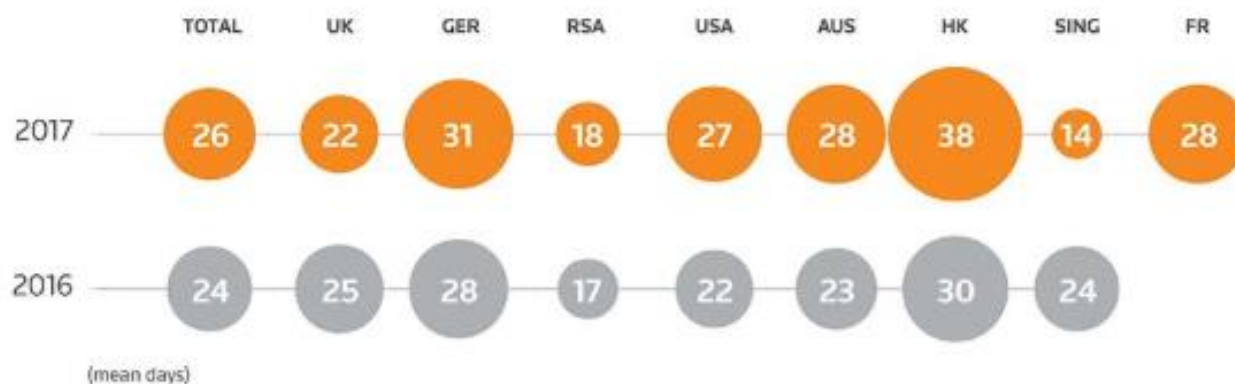
Poor-quality data, nonstandard data structures, and fragmented sources make data aggregation difficult. For example, many banks are still making tens of thousands of costly customer calls every month to refresh KYC documents, updating information that is incorrect or missing in their databases.

Costs of KYC are mainly driven by time spent on customer due diligence analysis



Fragmented systems and platforms limit the ability to automate transaction monitoring and due diligence. Instead, compliance teams spend the bulk of their time collecting data, and then on “stare and compare” sessions, instead of investigative work

KYC time during customer onboarding process increased to 30+ days by the end of 2017



KYC request between 2 banks might take even longer: 30 – 50 days using standard industry measures

Main pain points, both of which negatively impact client experience

- waiting time it takes to onboard client
- number of times the client is contacted for documentation (on average 4 contacts for private client and 8 for corporate customers)

Possible reasons

- external requirements (for instance requirement to provide hard copies of documents in Hong Kong)
- internal banks' internal processes (multiple teams conducting different aspects of the onboarding process leading to duplication of efforts further damaging client service standards and relationships).

One way damage could be reduced would be by minimizing the dependency on clients to provide documentation, instead of collecting as much information as possible from official sources in the public domain

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Areas of improvement

- automation of processes (such as investigator visualization tools)
- advanced statistical modeling (such as machine learning–based risk scoring and alert-generation engines)
- **efficient shared data-aggregation platforms**



	1 Central DB	Blockchain	
		2 Private	3 Public
• removes the duplication of effort in KYC checks	+	+	+
• enables updates to client details to be distributed to all banks in near real-time	+	+	+
• provides a historical record of all documents shared and compliance activities undertaken for each client		+	+
• ensures no central party could validate or change data			+

Goals



Reduce time of effective KYC due diligence and ensure a positive onboarding experience.



Automate customer risk and re-classification to focus on high-risk cases.



Automatically determine the risk rating of potential customers.



Quickly address changing regulations with business rules for risk levels and due diligence.

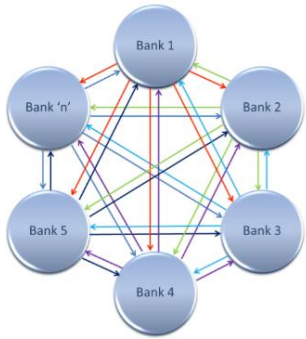


Improve the integrity of customer data and gain a holistic view on the customer.

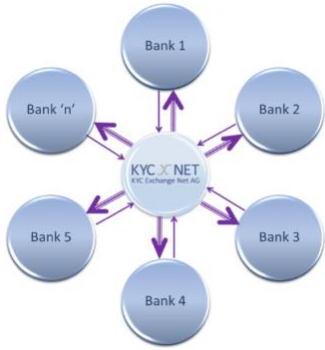


Easily integrate a robust standard application with open and flexible architecture.

1 KYC solutions based on centralized ledgers are widely adopted since 2014



- Most banks use internally designed KYC questionnaires when they initiate the KYC process, and **exchange the questionnaire by traditional email in pdf format.**
- This highly manual process results in **high fixed costs**, which affect the bank which requires KYC information from its clients, but also the clients who have to provide their information to a multitude of banking partners
- When this work is carried out by each institution individually the **effort is costly and duplicative**

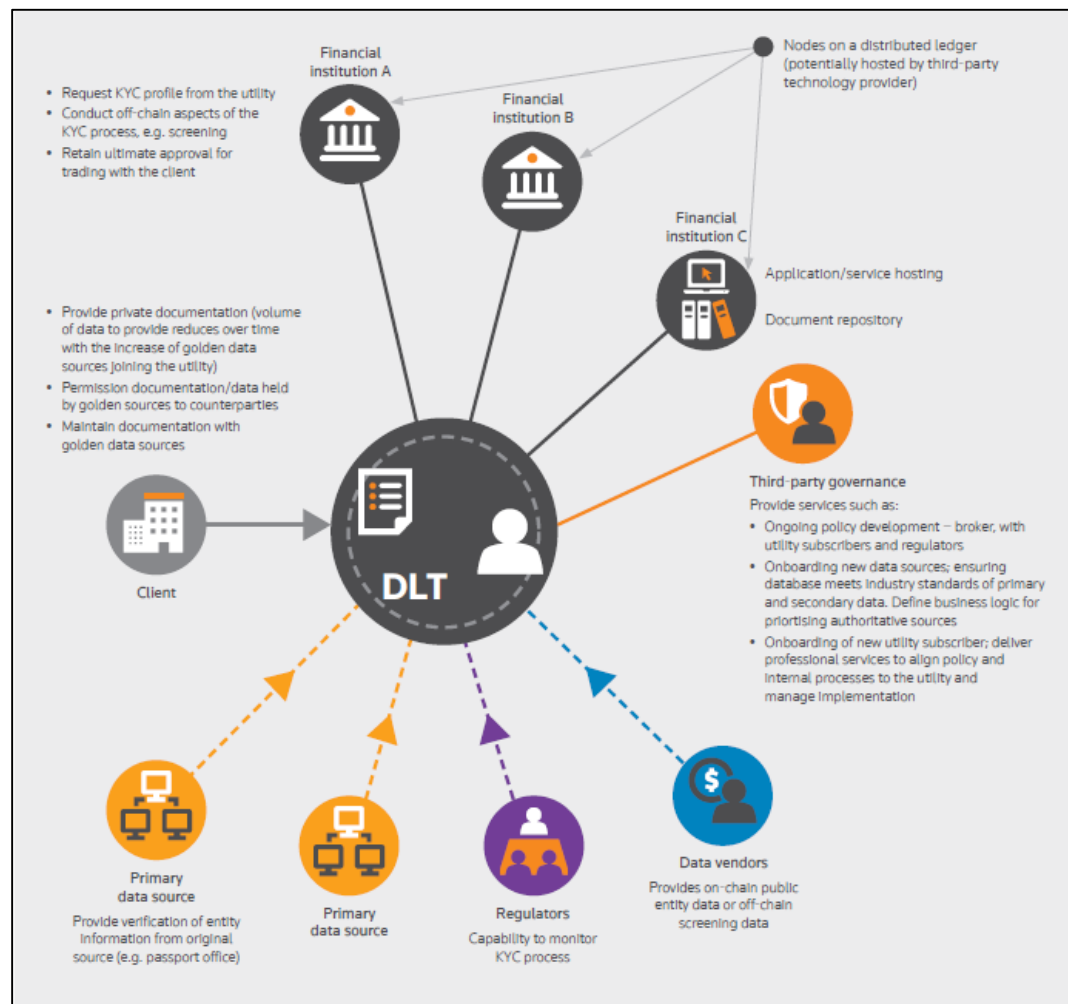


- **Shared KYC Registry addresses this challenge** by offering a single, secure global source of standardized KYC information
- In addition, if a user makes a change to any piece of information, such as an address, the system will keep track of which banks held the old information and will **send an automatic push update to notify them of the change**
- This has the added benefit of making it impossible to give conflicting information to different institutions, cutting down on the opportunities for cheating

- **SWIFT** launched the **SWIFT KYC Registry** in 2014, 2000+ banks enrolled
- In **2017 Thomson Reuters** acquired Clariant (global KYC platform formerly owned by Barclays, Credit Suisse, Goldman Sachs, J.P. Morgan among others) and Avox (leading supplier of legal entity data, e.g. hierarchies and identifiers on financial entities globally) creating KYC Platform with 350,000+ KYC records covering 1.25m managed legal entities in over 140 countries
- **KYC Exchange** went live in 2014, 30+ banks enrolled, incl. Standard Chartered Bank (SCB), Société Générale, the Bank of London and the Middle East (BLME)
- **e-Residency** is a centralized beta KYC registry (27,000 people from 143 countries)) by Estonian government providing easy integration with blockchain platforms
- **None of these and other large KYC Registries uses Blockchain as a core**

2 In 2018 KYC solution built on R3 Corda (private blockchain) were tested in large scale

- 45 nodes in Microsoft Azure carried out over 300 transactions for **70 of the biggest financial institutions** incl. BNP Paribas, China Merchants Bank and Deutsche Bank
- **Banks were able to request access to customer KYC test data, whilst customers could approve requests and revoke access**
- Customers were also able to **update their data** which was then **automatically updated for all banks** with permission to access it
- The system **allows corporate customers to create and manage their own identities** including relevant documentation and then **grant permission to multiple participants to access this data**
- This **reduces duplication and costs** by eliminating the need for each institution to individually attest and update KYC records. And, because only those with a need to see data have access to it, there are **no data privacy and security issues**



The Corda platform solves most of the KYC requirements, is trustless, allows controlling own data and supports gradual network adoption

3 Early-stage KYC solutions are being built on public blockchain, none is adopted yet

Singaporean government

In 2018 KPMG, three major banks and regulator in Singapore developed a proof-of-concept (POC) KYC utility on a blockchain platform. The prototype successfully passed the Monetary Authority of Singapore's test scenarios

Bankchain

Announced in 2018 by State Bank of India (India's largest bank). 30-member blockchain consortium of public and private sector banks in India, also Federal Bank, Deutsche Bank and UAE Exchange
The active projects for 2017 include shared KYC / AML, syndication of loans / consortium lending, trade finance, asset registry & asset re-hypothecation, secure documents, cross border payments, peer-to-peer payments, and blockchain security controls

IBM

IBM announced the successful completion of the Phase I Blockchain-based Shared Corporate KYC project with Deutsche Bank, HSBC, Mitsubishi Financial Group and the Treasuries of Cargill

Tradle

U.S. startup Tradle was set up in 2014. The company simplifies and secures KYC requirements using blockchain technology. Tradle is unique in favouring user-centered utility

An open blockchain enabled KYC solution could be on the horizon, but it requires significant collaboration across industry participants and a number of hurdles to overcome

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